PROJECT PROPOSAL

AI Based Historical Handwriting Recognition and Digital Preservation

1. Introduction

Historical documents contain a lot of important information about the past including peoples lives family history property records and social conditions.

Many of these documents are very old and difficult to read because of faded ink damaged pages and different handwriting styles.

The R.O.A.D. programme in Barbados is working to preserve historical records from the 18th and 19th century. These records include wills deeds and other legal documents that contain important historical information.

The problem is that even when these documents are scanned and saved digitally their text is still difficult to search because it is handwritten.

This project will focus on using AI and deep learning to read historical handwritten text from images and convert it into digital text that can be searched and used more easily.

2. Problem Statement

Reading old handwritten documents is a difficult and time consuming task. A person may need to read each document manually and type the text which takes a lot of effort.

Historical handwriting is also different from modern handwriting. Letters may have unusual shapes and some words may be written in old spelling styles.

Some images also contain faded ink and damaged areas which makes it harder for a model to recognise the correct text.

The main problem is to develop a machine learning model that can recognise historical handwritten text with good accuracy even when the images have different writing styles and quality issues.

3. Project Aim

The main aim of this project is to develop a deep learning model that can automatically recognise and transcribe historical handwritten text from scanned images.

The project will use the Barbados Historic Handwriting dataset available through the R.O.A.D. challenge on Zindi.

The long term goal is to explore how this technology can help make historical records easier to read search and preserve for future use.

4. Objectives

The main objectives of this project are.

1. Study the dataset and understand the main problems in recognising historical handwriting.

2. Develop a basic handwriting recognition model and measure its initial performance.

3. Test pretrained models and fine tune them using the available historical handwriting data.

4. Try different image preprocessing and data augmentation techniques to improve the results.

5. Compare different model settings and find which methods work better.

6. Evaluate the model using Character Error Rate and Word Error Rate.

7. Study the common mistakes made by the model such as confusing similar letters or reading old words incorrectly.

8. Develop a simple prototype that takes a handwritten image and returns the predicted text.

5. Proposed Methodology

The project will be completed in different stages starting from dataset analysis and moving towards model training testing and prototype development.

Stage 1. Dataset Analysis

The first step will be to study the dataset provided for the competition. It contains around 6000 cropped images of handwritten text lines from historical documents.

I will examine the image quality handwriting styles image sizes and available text labels. This will help me understand the main challenges before selecting a suitable model.

Stage 2. Data Preprocessing

Historical documents may contain faded writing noise and uneven backgrounds. Different preprocessing techniques will be tested to see if they help the model recognise the text more accurately.

These techniques may include contrast adjustment resizing normalisation and limited noise reduction. Care will be taken to avoid removing important details from the handwriting.

Data augmentation will also be tested by making small changes in brightness contrast and blur. This may help the model work better with images that have different visual conditions.

Stage 3. Model Selection and Training

Instead of training a large model from the beginning I plan to investigate pretrained handwriting recognition models and fine tune them on the available dataset.

Possible approaches include transformer based models and models that combine convolutional neural networks with sequence recognition methods.

A baseline model will first be developed to measure the initial results. After that different approaches will be tested to improve the accuracy.

Stage 4. Model Improvement

After getting the baseline results I will experiment with different training settings and techniques.

This may include trying different image sizes augmentation methods learning rates and decoding strategies. The purpose will be to understand which changes actually improve the results.

The model will not be made more complex without a clear reason. The focus will remain on improving recognition accuracy while keeping the system practical to train and use.

Stage 5. Evaluation

The model will be evaluated using Character Error Rate and Word Error Rate.

CER measures the errors at character level while WER measures errors at word level. Both are important because even small spelling mistakes can affect the quality of a historical transcription.

The improved model will be compared with the baseline and common errors will be studied. These may include missing characters confusing similar letters incorrect spacing and difficulty reading faded handwriting.

The validation data will be used to select the model and its settings. The test data will be reserved for final evaluation according to the competition rules.

Stage 6. Prototype Development

After developing a suitable model I plan to create a simple prototype where a user can provide an image of handwritten text and receive its predicted transcription.

This will demonstrate how the model could be used in a practical application.

In the future the same idea could be extended to process larger collections of historical documents and make their contents searchable.

6. Proposed System

The system will follow a simple workflow.

Historical Document Image

Image Preprocessing

Handwriting Recognition Model

Text Decoding

Digital Text Output

Each stage will be tested separately where possible. This will make it easier to identify problems and improve the system.

In the future the system could also include features such as searching historical records identifying names and dates and extracting useful information from documents.

These features will be considered future work because the first priority is to develop a reliable handwriting recognition model.

7. Expected Outcomes

By the end of the project I expect to develop a working deep learning model for recognising historical handwritten text.

The project will include a baseline model and improved versions for comparison. Different training and preprocessing techniques will also be evaluated.

The results will be measured using CER and WER and the common errors made by the model will be analysed.

A simple prototype will also be developed to convert handwritten images into digital text. The final report will explain the methods experiments results and limitations of the project.

The main target will be to improve transcription accuracy over the baseline model. The actual improvement will depend on the dataset and experimental results.

8. Importance of the Project

This project is useful because historical archives contain information that can be difficult to access through manual reading alone.

From a technical point of view it will provide an opportunity to study how deep learning and computer vision can solve a real world problem.

From a practical point of view automated transcription could reduce the amount of manual work required to make historical records searchable. Researchers and historians could find information more easily without reading every scanned page manually.

The same approach may also be useful for other historical archives although different handwriting styles and datasets would require further testing.

The model may not recognise every word correctly especially when the handwriting is unclear or damaged. Human checking will still be important for names dates and other important historical details.

9. Conclusion

This project aims to develop an AI based system for recognising and transcribing historical handwritten text from scanned images.

Using the Barbados Historic Handwriting dataset I will investigate different deep learning models and training techniques to improve the accuracy of historical text recognition.

The project will start with a basic model and then focus on improving its performance through experiments and evaluation.

The final outcome will be a trained model a comparison of results and a simple prototype for handwritten text transcription.

In the longer term this work could help make historical archives easier to search and study while providing a starting point for more advanced AI systems that can help preserve historical information for future generations.